

Applied Optimization

Exercise 10

Inequality Constrained Optimization

Active set method for convex quadratic problem (0.75 pt)

Consider the following 2-dimensional minimization problem

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 \\ \text{subject to:} \quad & x-2y+2 \geq 0, \quad (C_1) \\ & -x-2y+6 \geq 0, \quad (C_2) \\ & -x+2y+2 \geq 0, \quad (C_3) \\ & x \geq 0, \quad (C_4) \\ & y \geq 0 \quad (C_5) \end{aligned}$$

use the active set method for convex QPs starting from the following points:

1. $(x^0, y^0) = (1, 1/2)^T$ for $\mathcal{A}^0 = \emptyset$
2. $(x^0, y^0) = (3, 1/2)^T$ for $\mathcal{A}^0 = \{3\}$
3. $(x^0, y^0) = (0, 1/2)^T$ for $\mathcal{A}^0 = \{4\}$

Sketch the feasible set and for every starting point:

1. Sketch the path of the active set method.
2. Verify the path of the active set method computationally.
 Note for each step the following states in the table below:

- the primal variables before the step (x_k, y_k)
- the active set before the step A_k
- the primal variables after the step (x_{k+1}, y_{k+1})
- the dual variables after the step λ_{k+1} , e.g. $(0, 1, -1, 0, 0)$
- the active set after the step A_{k+1}
- whether the KKT conditions are satisfied for $(x_{k+1}, y_{k+1}, \lambda_{k+1})$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	$(1, 1/2)$	$\emptyset$				
1						
2						
3						

[Hint: See Nocedal example 16.4 p. 475. and exercise 16.10]

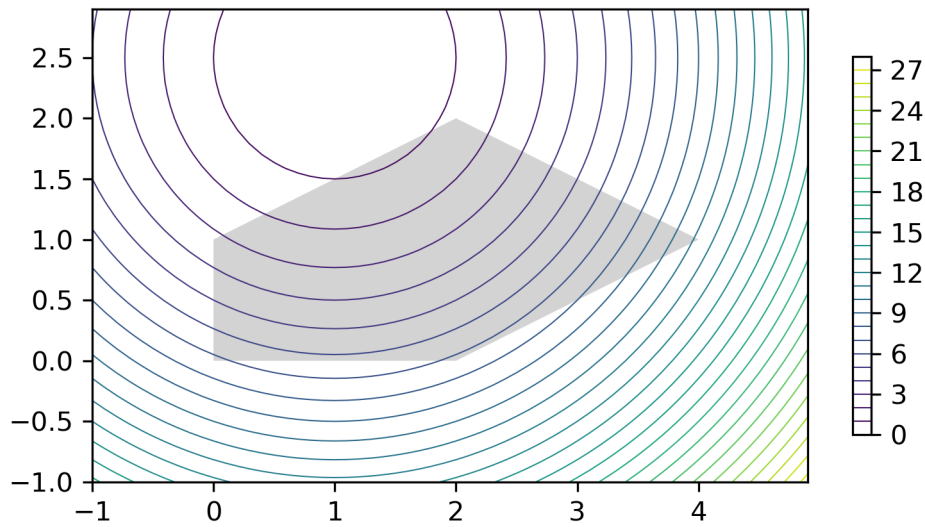


Figure 1: Problem: objective function iso-lines and feasible set.

Solution:

Figure 1 depicts the problem: objective function as iso-lines and shaded area as feasible set.

We use the algorithm presented in the lecture slides, repeated below for convenience.

The problem data is summarized as follows, for the objective function:

$$Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad q = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$$

Rewrite constraints in standard form:

$$\begin{pmatrix} -1 & 2 \\ 1 & 2 \\ 1 & -2 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \preceq \begin{pmatrix} 2 \\ 6 \\ 2 \\ 0 \\ 0 \end{pmatrix},$$

Algorithm: Active-Set Method for Convex QP

Input : feasible starting point x_0 , and initial active set A_0

Output: approximate x^*, λ^*

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for  $k = 0, 1, 2, \dots$  do
    /* Solve KKT System with constraints  $A_k$  to obtain  $x_{k+1},$ 
        $\lambda_{k+1}$  */
    
$$\begin{bmatrix} Q & B_k^T \\ B_k & 0 \end{bmatrix} \begin{bmatrix} x_{k+1} \\ \lambda_{k+1} \end{bmatrix} = \begin{bmatrix} -q \\ b_k \end{bmatrix}$$

    /* Check whether movement impossible */
    if  $x_{k+1} = x_k$  then
        if  $\lambda_i \geq 0$  for all  $i \in A_k$  then
            stop with solution  $x^* = x_{k+1}$ 
        else
            /* Remove constraint with most negative  $\lambda$  */
             $j = \arg \min_{i \in A_k} \lambda_{k+1}$ 
             $A_{k+1} = A_k - \{j\}$ 
             $x_{k+1} = x_k$ 
        end
    else
        if there are blocking constraints in inactive constraint set  $\bar{A}$  then
            /* Add first blocking constraint */
             $j = \arg \min_{j \in \bar{A}} (\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)})$ 
             $x_{k+1} = x_k + \alpha_j (x_{k+1} - x_k)$ 
             $A_{k+1} = A_k \cup \{j\}$ 
        else
            /* Do full step */
             $A_{k+1} = A_k$ 
        end
    end
end
end
```

1. $(x^0, y^0) = (1, 1/2)^T$ and $A_0 = \emptyset$:

$k=0$: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix}$, but C1 is blocking $\rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1.5 \end{pmatrix}$ and we add C1 to the active set.

$k=1$: $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 1.7 \\ 0.8 \end{pmatrix}$

x hasn't changed and $\lambda_1 > 0 \rightarrow$ we can stop here.

Table 1 summarizes the computations for this starting point and figure 2 depicts this path.

Table 1: Solution sequence for $(x^0, y^0) = (1, 1/2)^T$ for $A_0 = \emptyset$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(1.0, 0.5)	$\emptyset$	(1.0, 1.5)	(0, 0, 0, 0, 0)	{1}	No
1	(1.0, 1.5)	{1}	(1.40, 1.70)	(0.8, 0, 0, 0, 0)	{1}	Yes

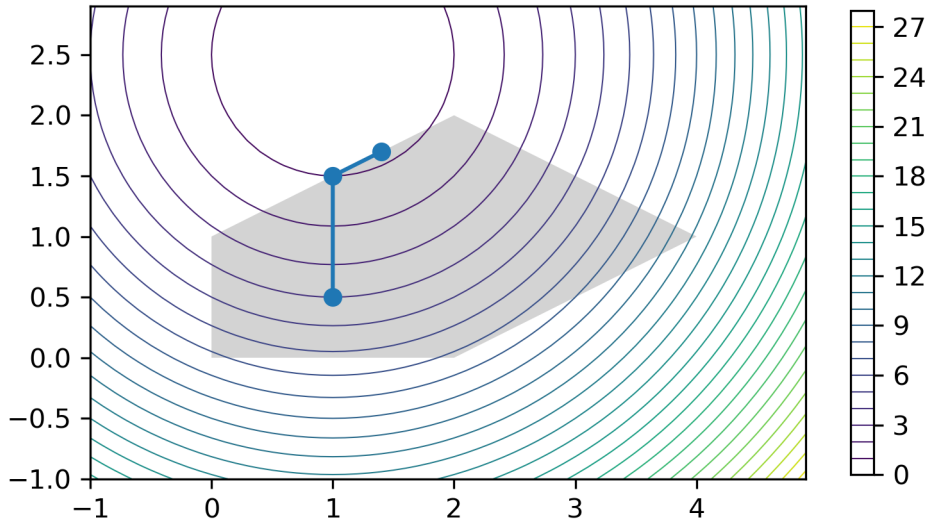


Figure 2: Path 1

2. $(x^0, y^0) = (3, 0.5)^T$ for $A_0 = \{3\}$:

$$\underline{k=0}: \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & -2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} 2.2 \\ 0.1 \\ -2.4 \end{pmatrix}$$

$\underline{k=1}$: We solve the same system, thus get the same next position and thus remove C3 from the active set.

$$\underline{k=2}: \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix}, \text{ but C1 is blocking} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2.2 \\ 0.1 \end{pmatrix} + \alpha \begin{pmatrix} 1 - 2.2 \\ 2.5 - 0.1 \end{pmatrix} \text{ satisfies C1. So } \alpha = 2/3, \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1.4 \\ 1.7 \end{pmatrix}$$

$$\underline{k=3}: \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 1.7 \\ 0.8 \end{pmatrix}$$

Table 2 summarizes the computations for this starting point, and Figure 3 plots this path.

Table 2: $(x^0, y^0) = (3, 0.5)^T$ for $A_0 = \{3\}$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(3.0, 0.5)	{3}	(2.20, 0.10)	(0, 0, -2.4, 0, 0)	{3}	No
1	(2.20, 0.10)	{3}	(2.20, 0.10)	(0, 0, -2.4, 0, 0)	$\emptyset$	No
2	(2.2, 0.1)	$\emptyset$	(1.4, 1.7)	(0, 0, 0, 0, 0)	{1}	No
3	(1.4, 1.7)	{1}	(1.4, 1.7)	(0.8, 0, 0, 0, 0)	{1}	Yes

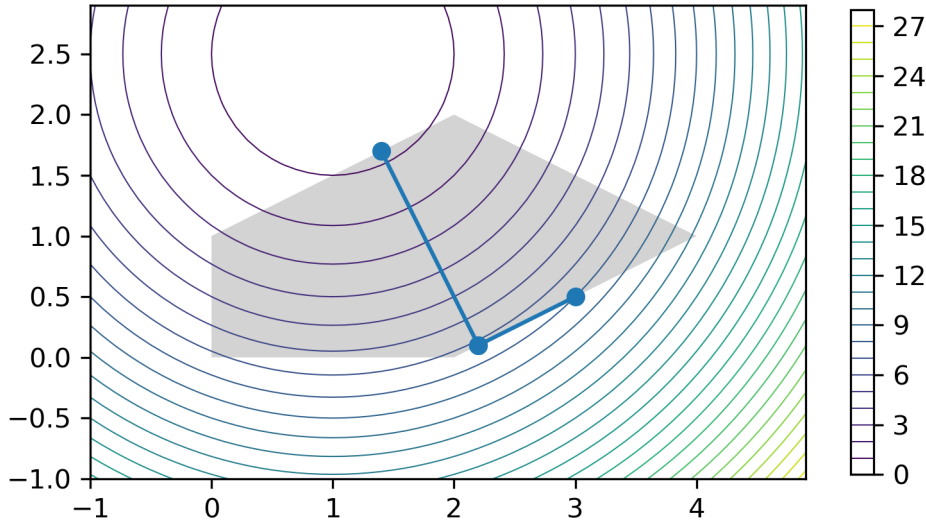


Figure 3: Path 2

3. $(x^0, y^0) = (0, 1/2)^T$ and $A_0 = \{4\}$

$\underline{k=0}$: $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 2.5 \\ -2 \end{pmatrix}$, but C1 is blocking $\rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} + \alpha \begin{pmatrix} 0-0 \\ 2.5-0.5 \end{pmatrix}$ satisfies C1. So $\alpha = 1/4$, $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
Add C1 to the active set.

$\underline{k=1}$: $\begin{pmatrix} 2 & 0 & -1 & -1 \\ 0 & 2 & 0 & 2 \\ -1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_4 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 0 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_4 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -3.5 \\ 1.5 \end{pmatrix}$ Remove

C4 from the active set.

$\underline{k=2}$: $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 1.7 \\ 0.8 \end{pmatrix}$

Table 3 summarizes the computations for this starting point. Figure 4 depicts this path.

Table 3: $x^0 = (0, 1/2)^T$ for $A_0 = \{4\}$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(0, 0.5)	$\{4\}$	(0, 1)	(0,0,0,-2,0)	$\{1,4\}$	No
1	(0, 1)	$\{1,4\}$	(0, 1)	(1.5,0,0,-3.5,0)	$\{1\}$	No
2	(0, 1)	$\{1\}$	(1.40, 1.70)	(0.8,0,0,0,0)	$\{1\}$	Yes

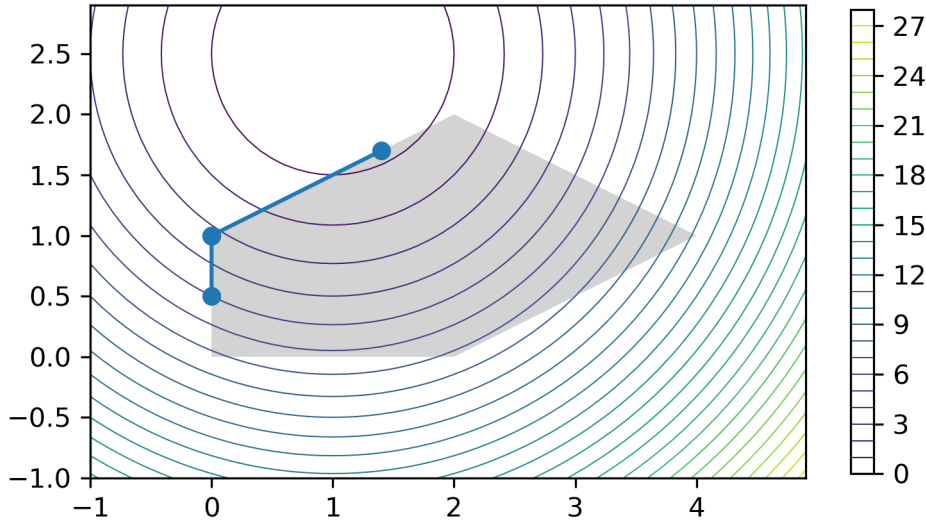


Figure 4: Path 3

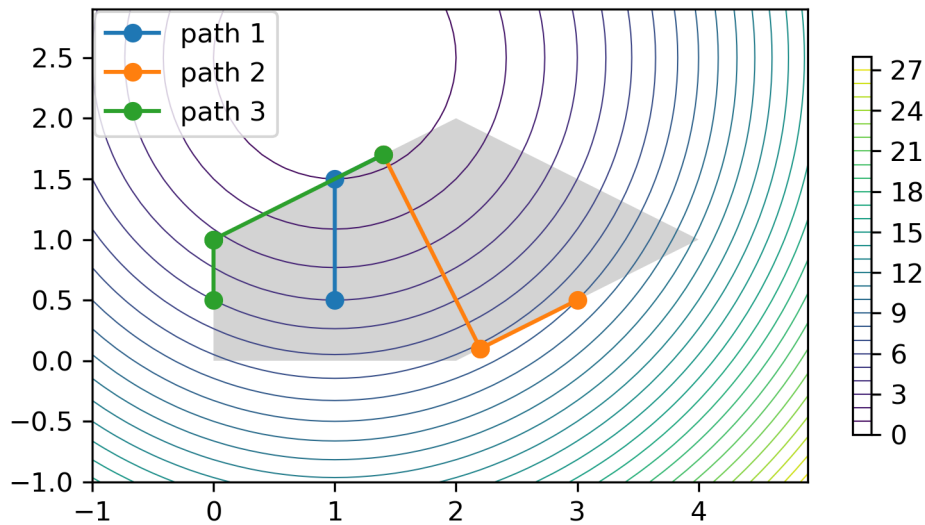


Figure 5: Optimization paths corresponding to the three given cases, all have the same endpoint, overlapped in the figure.

"Phase I" method for initial feasible point (0.25 pt)

The active set method used in the task above requires an initial feasible point. Use the "Phase I" approach as described in the lecture and the book [Nocedal, Section 16.5] to compute an initial feasible point for the problem above, starting from $x = (1, 2.5)^T$. You can use a computer algebra system if needed. [Hint: formulate the "Phase I" problem by adding slack variables for violated constraints at the start point. Solve the active set subproblem with active set method or gradient descent (projected newton) in combination with constraint elimination].

Solution:

Starting from infeasible point $\tilde{x} = (1, 2.5)^T$, we observe that only constraint 1 is violated, thus our "Phase I" feasibility problem is:

$$\begin{aligned} \min \quad & z \\ \text{subject to:} \quad & -x + 2y - 2 \leq z, \\ & x + 2y - 6 \leq 0, \\ & x - 2y - 2 \leq 0, \\ & -x \leq 0, \\ & -y \leq 0 \\ & z \geq 0 \end{aligned}$$

we can compute a feasible starting point for the given $\tilde{x} = (1.0, 2.5)^T$, the problem is feasible with the following $\tilde{z}$:

$$\tilde{z} = \max(-1 + 2 \times 2.5 - 2, 0) = \max(4 - 2, 0) = \max(2, 0) = 2$$

Active set method:

Notice that the "Phase I" feasibility problem has the same optimum as $z \geq 0$:

$$\begin{aligned} \min \quad & 1/2z^2 + z \\ \text{subject to:} \quad & -x + 2y - 2 \leq z, \\ & x + 2y - 6 \leq 0, \\ & x - 2y - 2 \leq 0, \\ & -x \leq 0, \\ & -y \leq 0 \\ & z \geq 0 \end{aligned}$$

$$\underline{k=0}: \begin{pmatrix} 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \\ 6 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ z \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 1.75 \\ -1 \\ 0 \\ 0 \end{pmatrix},$$

but $-z \leq 0$ is blocking $\rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} 2.5 - 1 \\ 1.75 - 2.5 \\ -1 - 2 \end{pmatrix}$ satisfies $-z \leq 0$. So

$\alpha = 2/3$, $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$ Add $-z \leq 0$ to the active set.

$$\underline{k=1}: \begin{pmatrix} 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ \lambda_1 \\ \lambda_2 \\ \lambda_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \\ 6 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ z \\ \lambda_1 \\ \lambda_2 \\ \lambda_6 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

The KKT conditions are satisfied and our optimum is 0 with optimum solution $(x^*, y^*, z^*) = (2, 2, 0)$. Since $z = 0$, the point $(x, y) = (2, 2)$ is a feasible point of the original problem.

Gradient Descent:

One way to solve the above problem from the feasible starting point $(\tilde{x}, \tilde{y})^T = (1, 2.5, 2.5)^T$ is via variable elimination and then perform a gradient descent step with a step length that is computed from the constraints as follows. The resulting active set problem is then:

$$\begin{aligned} \min \quad & z & (1) \\ \text{subject to:} \quad & -x + 2y - 2 - z = 0 & (2) \\ & x + 2y - 6 = 0 & (3) \end{aligned}$$

From (3):

$$x = 6 - 2y \quad (4)$$

(4) in (2):

$$\begin{aligned} -(6 - 2y) + 2y - 2 - z &= 0 \\ z &= 4y - 8 \quad (5) \end{aligned}$$

(5) in (1) we get the new objective function

$$f(z) = 4y - 8$$

whose gradient is $\nabla f = 4$. Thus the gradient descent update for the complete variable set (x, y, z) from equations (4) and (5) is:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 - 2(2.5 - 4t) \\ 2.5 - 4t \\ 4(2.5 - 4t) - 8 \end{pmatrix} = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix} - t \begin{pmatrix} -8 \\ 4 \\ 16 \end{pmatrix}$$

The step t can be as long as the inequality constraints are not violated: in this case we first ask $x, y, z \geq 0$. Thus:

$$2.5 - 4t \geq 0 \Rightarrow t \leq 5/8$$

and

$$2 - 16t \geq 0 \Rightarrow t \leq 1/8$$

We try $t = 1/8$ and the other inequality constraints are satisfied too. So the solution is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}^* = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix} - \frac{1}{8} \begin{pmatrix} -8 \\ 4 \\ 16 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$$

Since $z = 0$, the point $(2, 2)$ is a feasible point of the original problem.

Problem 3: "Big M" method for initial feasible point (Bonus 0.5 pt)

Consider now the "Big M" method and use it to compute the initial point also starting from $x = (1, 2.5)^T$ with $M = 100$.

Solution:

We start by adding a penalty term $M\eta$ to the objective function, and $\eta = \max(g_i(x), 0)$ to get a feasible point of the following problem:

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 + M\eta \\ \text{subject to:} \quad & g_1 = -x + 2y - 2 \leq \eta, \\ & g_2 = x + 2y - 6 \leq \eta, \\ & g_3 = x - 2y - 2 \geq \eta, \\ & g_4 = -x \leq \eta, \\ & g_5 = -y \leq \eta, \\ & g_6 = \eta \geq 0 \end{aligned}$$

At the given $(\tilde{x}, \tilde{y}) = (1, 2.5)$, g_1 is the maximum and thus $\eta = 2$. Following the idea of active set method, the only active constraint is C_1 , and we can convert the problem to the equality constrained problem:

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 + 100\eta \\ \text{subject to:} \quad & -x + 2y - 2 = \eta \end{aligned}$$

We can proceed by variable elimination, and compute the unconstrained minimizer,

$$f(x, y) = (x-1)^2 + (y-2.5)^2 + 100(-x + 2y - 2)$$

The gradient at the start point $(x, y, \eta) = (1, 2.5, 2)$ is

$$\nabla f(x, y) = \begin{pmatrix} 2(x-1) - 100 \\ 2(y-2.5) + 200 \end{pmatrix} = \begin{pmatrix} -100 \\ 200 \end{pmatrix}$$

Assuming the step length as t , the next point $\begin{pmatrix} 1 - (-100t) \\ 2.5 - 200t \end{pmatrix} = \begin{pmatrix} 1 + 100t \\ 2.5 - 200t \end{pmatrix}$

Put the new point back to the equality constraint and we get $-(1 + 100t) + 2(2.5 - 200t) - 2 = 2 - 500t = \eta \geq 0$.

To get a feasible point of the original problem, we ask $\eta = 0$, thus $t = 2/500$. We try $t = 2/500$ and the point $(x, y) = (1.4, 1.7)$ satisfies the constraints $C_2 - C_6$ too.